

Solve θ , d , a , α from endpoint_vector and rotation_vector:

Assume:

$$\vec{p} = (x_p, y_p, z_p) \quad \text{endpoint_vector}$$

$$\vec{r} = (x_r, y_r, z_r) \quad \text{rotation_vector}$$

$$O_0 = (0, 0, d) = \vec{z}$$

$$O_1 = \vec{p} + \lambda \vec{r} = (x_p + \lambda x_r, y_p + \lambda y_r, z_p + \lambda z_r)$$

$$\overrightarrow{O_0 O_1} = (x_p + \lambda x_r, y_p + \lambda y_r, z_p + \lambda z_r - d)$$

$$\because \overrightarrow{O_0 O_1} \perp \vec{r}, \quad \overrightarrow{O_0 O_1} \perp \vec{z}$$

$$\therefore \begin{cases} \overrightarrow{O_0 O_1} \cdot \vec{r} = 0 \\ \overrightarrow{O_0 O_1} \cdot \vec{z} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} (x_p + \lambda x_r)x_r + (y_p + \lambda y_r)y_r + (z_p + \lambda z_r - d)z_r = 0 \\ (z_p + \lambda z_r - d)d = 0 \end{cases}$$

case 1: $d = 0$

$$\begin{cases} \lambda = -\frac{x_p x_r + y_p y_r + z_p z_r}{x_r^2 + y_r^2 + z_r^2} \\ d = 0 \end{cases}$$

case 2 $(z_p + \lambda z_r - d) = 0 \Rightarrow d = z_p + \lambda z_r$

$$\begin{cases} \lambda = -\frac{x_p x_r + y_p y_r}{x_r^2 + y_r^2} \\ d = z_p - \frac{x_p x_r + y_p y_r}{x_r^2 + y_r^2} \cdot z_r \end{cases}$$

$$\therefore a = \sqrt{(x_p + \lambda x_r)^2 + (y_p + \lambda y_r)^2 + (z_p + \lambda z_r - d)^2}$$

$$\vec{X} = (1, 0, 0), \quad \vec{Z} = (0, 0, 1)$$

$$\theta = \angle(\vec{X}, \overrightarrow{O_0O_1})$$

$$\alpha = \angle(\vec{Z}, \overrightarrow{O_0O_1})$$